Circular Saw Mechanical Failure Analysis

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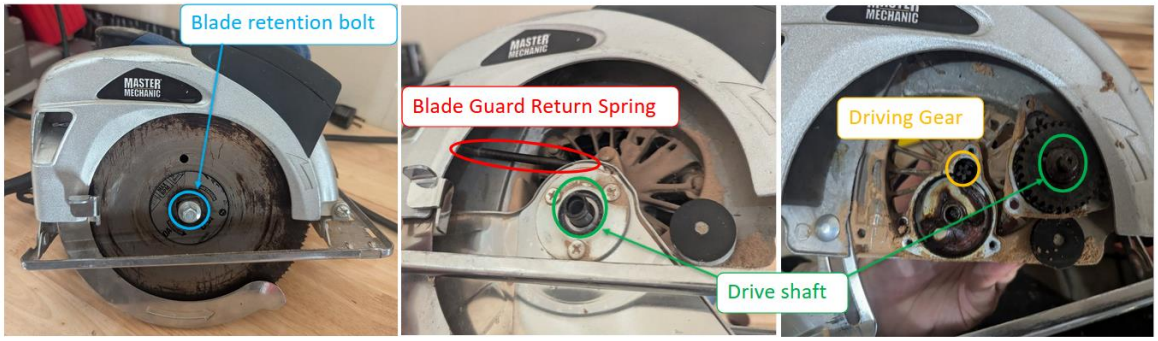
Introduction:

In this lab, we aim to perform a mechanical design analysis on a 7 ¼ corded circular saw, and in particular various components contained within said saw. A circular saw is a power tool designed for cutting commonly used materials such as wood, metal, and plastic. This tool works by using a wall outlet to supply power to a motor that drives a circular blade through a **driving gear** system. The blade is secured to the **drive shaft** by means of a **blade retention bolt**. There is also a **spring actuated** blade guard included to mitigate the severe bodily harm the tool can cause, especially in the case of a kickback event. By examining the stresses acting on critical parts of the saw, we can better understand the mechanical components and forces that act on it during operation. For this analysis, each group member will focus on a specific part of the saw: the driving gear, drive shaft, blade guard return spring, and blade retention bolt.

Product:	7 ¼" Corded Circular Saw with Blade Guard
Customer type:	DIYers, hobbyists
Market niche:	Low cost, low usage frequency
Key considerations:	Price, safety, ease of use

Master Mechanic 7-¼" Corded Circular Saw

The Saw Disassembled



The individually analyzed components located on the saw.

Design case: Cross-cutting a 4x4 beam of Douglas Fir wood.

Reasoning:
A 4x4 is 3.5in thick on each side. This is more than what a 7-¼ saw would be able to cut (meaning the blade would not cut all the way through from one full-depth cut).
This is a very common scenario a user might encounter and is a good example of the blade being used at its limit.

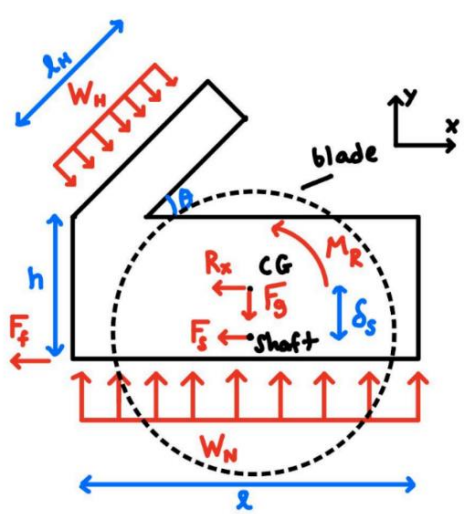
Example of the design case. The pictured wood is pressure treated. (Picture source)

To simplify the analysis for the entire saw, several assumptions have been made. We will treat the saw as a simple box, with its center of mass located at the geometric center. The resultant forces from saw operation will be simplified as acting on the shaft. Additionally, the handle of the saw will be modeled as a one-dimensional line, considering only its length. We will also be omitting the force exerted by the blade guard against the work piece (the force that results from the spring being stretched.)

Most dimensions and variables were measured. The distributed load exerted on the handle was estimated by pushing against a scale with a force expected from muscle memory and dividing this force by the handle length. The force on the shaft was estimated based upon the x-component of the force applied on the handle—some rudimentary calculations were performed (ie, treating the FBD as a dynamic scenario, where R_x is the resulting force).

Note: All referenced figures, equations and tables in the overall saw analysis are from *Shigley's Mechanical Engineering Design 10th Edition*.

Free body diagram and force analysis results: (See spreadsheet for calculations.)



Free body diagram of the corded circular saw in the chosen design scenario

Variable	Description	Value	Units
ℓ_h	Handle length	6	in
W_h	Hand distributed load	3.33	lb-f/in
θ	Handle angle	45	degrees
F_g	Saw weight	9	lb-f
F_s	Force acting on shaft	5	lb-f
h	Height of saw body	5	in
F_f	Friction force	6.36	lb-f
W_N	Normal distributed force	12.72	lb-f/in
ℓ	Length of saw body	10	in
δ_s	Shaft offset	1	in
R_x	Constraining force for static analysis	2.77	lb-f
M_R	Constraining moment for static analysis	45.51	in*lb-f

Critical feature	Customer need	Engineering Requirement
Secure blade attachment:	The blade must be fastened to the shaft, so it doesn't loosen or wiggle.	The blade must be fastened in a way to withstand all the torque on the saw to prevent it from any movement and keep it stable. This would require a minimum torque that it would need to withstand.
Quick trigger response:	The trigger should be instantaneous in controlling the saw blade to allow for an immediate stop and start.	The trigger would require a certain amount of force to push the trigger down and be able to disengage the saw very fast within 0.1 seconds.
Comfortable handle design:	The handle should be designed, so that it is easy to hold and allows the user to hold for longer periods without fatigue.	The handle must be designed to need a minimal amount of force to lift, hold, and use the saw while minimizing user fatigue.
Cutting speed:	Blade must be able to cut through common materials relatively quickly. (Lumber, sheet metal, plastics.)	Motor must provide sufficient torque and speed—based on customer needs for typical materials.
Kickback risk mitigation:	Saw must mitigate risk of kickback.	Safety controls and/or mechanisms must be in place to mitigate the inevitability of kickback for this type of saw.
Power supply:	Saw must not trip a breaker while under load while connected to a typical 120V-15A wall outlet.	Motor must be specified and geared to adequately meet current draw, torque, and speed needs for common materials within some margin while connected to 120V mains.
Blade size:	The blade must be within a reasonable size range for a person to use.	The blade should be greater than 2 inches and less than 4 feet in diameter.
Saw form factor:	Saw must be lightweight, compact design to reduce fatigue during extended use.	Optimized material selection and structural design should keep the saw's weight within the desired range (e.g., 4-10 pounds) to allow the customer to maintain portability without losing durability.
Blade durability:	The blade should be sturdy enough to minimize long term wear and tear.	The material of the blade should exceed standards in 4 measured metrics: hardness, toughness, heat resistance, and corrosion penetration rate
Blade replacement:	The blade should be able to be replaced without impacting the effectiveness of the saw.	The connection between the blade and the saw housing should be strong enough to withstand the torque created by the cutting, but should be simple enough for any saw owner to easily take out on their own.

Using these customer needs and engineering requirements, we filled out a House of Quality to determine the critical aspects to consider. From this HoQ, the Failure Modes and Effects Analysis, and the consideration of the typical customer profile for this specific tool, we determined that the components we chose to analyze were critical to the **safety** and/or **functionality** of the saw.

Failure Modes and Effect Analysis:

Name	Function	Failure Mode	Failure Effect	Severity	Causes	Occurence	Identification and mitigation	Detection	RPN
	Describe	Describe	Describe	10 = very bad 1 = trivial	Describe	10 = always 1 = never	Describe	10 = Impossible 1 = Obvious	
Bolts	Blade Mounting Bolt	Bolt Loosening	If the bolts were to loosen over time, the potential failure effect would be the blade detaching causing it to possibly fly off.	10	Vibrations causing bolts to loosen	3	Blade would be wobbly or come off. Mitigated by checking blade stability before use.	7	210
Bolts	Shaft that connects to saw blade	Thread Stripping	Stripped threads would prevent components from being fully fastened and make them misaligned. This could reduce cutting accuracy and damage the saw with further use.	6	Replacing or changing blades and tightening the bolts on too tight or with improper alignment could strip the screws	5	If the blade is loose or if the screws provide no resistance when being twisted the threads are most likely stripped. Avoid overtightening for the future.	3	90
Shaft	Shaft that connects to saw blade	Wear and tear from the high torques being put on the saw can cause deformation on the shaft of the saw	When the shaft of the saw deforms, it can cause wide inaccuracies within the saw. Since the metal gradually deforms, the inaccuracies wouldn't be immediately noticeable, but over time, greater deformation in the shaft would cause large inaccuracies in the cuts over time.	4	Could cause inaccuracies and difficulty in cutting for the user.	6	User will only notice once the shaft is already deformed. It won't be catastrophic by the time they notice some difficulty in cutting, but it might be hard to detect the exact reason why their saw is becoming less accurate over time.	6	144
Shaft	Shaft that connects to saw blade	Sudden extreme changes in torsion on the saw could wildly deform(or even break) the shaft.	Having too much torsion on the shaft could have extremely dangerous impacts on both the saw and the person handling the saw. A broken shaft would cause extreme initial kickback which could seriously injure the person and would render the saw useless.	10	Could cause serious injury to the user and complete breakage of the saw	2	Extremely obvious when this happens. It is basically impossible to miss.	1	20
Springs	Trigger Return: Returns the trigger to the off position	Springs in saws can fail due to being compressed and released over time due to repeated use.	When the springs in the trigger return fail, it could cause unintended operation after trigger release. This effect could make the customer lose control over the saw, which might result in unintended cuts, damage to material, or personal injury.	8	Repeated use of the trigger	5	If tool doesn't run when trigger is pressed or doesn't stop when released	1	40
Springs	Blade guard return spring mechanism	Corrosion of the springs	If the spring mechanism is exposed or not properly maintained, then the corroded spring in the blade guard may lose its tension, causing the guard to malfunction. The guard may fail to retract or extend properly, leaving the spinning blade exposed, significantly increasing the risk of injury.	10	When the spring is exposed to the elements	5	Blade guard is not returing to the original position leaving the blade exposed.	1	50
Gears	Gear attached to motor output shaft	Gear Teeth shear off	Motor does not transmit power to the blade - blade can no longer operate.	10	Misuse, sudden jams, or overloading the saw	3	The saw will no longer be able to rotate or spin	7	210
Gears	Gear attached to arbor shaft	Wear and tear of gears through repeated rigourous use	Reduced cutting performance and gear slippage which deteriorates the saw until the blade is no longer able to spin.	7	Continuous use over time and any caused misalignment or lack of lubrication.	2	Excessive vibration or whining of gears would be key identify factors. This can be mitigated by consistent lubrication and not overloading the saw.	9	126

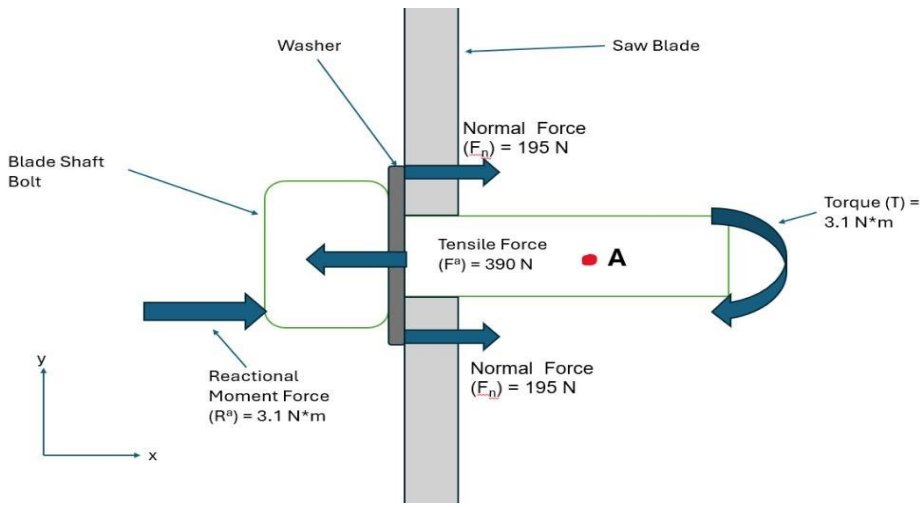
Component	Importance for Safety	Importance for Functionality	Justification for Analysis
Blade retention bolt	High	High	This bolt is the only thing holding the blade onto the drive shaft. This is one of the most important components in regard to safety and functionality.
Driving gear	Minimal	High	The driving gear is the primary mechanism through which the motor power is transmitted to the blade. Critical to function, but failure has little impact on safety.
Blade guard return spring	High	Minimal	This spring ensures that the blade safety guard is engaged immediately after making a cut. Failure of this component is trivial in terms of functionality, but has dire consequence for safety.
Drive shaft	High	High	This shaft transmits the power generated from the motor directly to the blade. Without this component, the blade will not spin and a total failure could cause the blade to come loose which is extremely dangerous.

Components Failure Analysis:

Blade Retention Bolt:

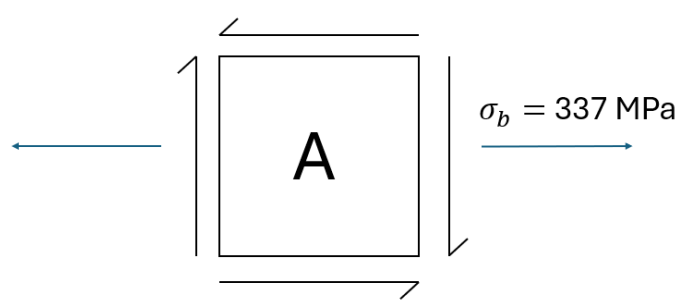
Below is the free body diagram for the blade retention bolt with both the tensile force and torque acting on it, but with the assumption that for the stress on the actual bolt, the tensile force is all that matters as the torque is on the shaft and causes the tensile force. Other assumptions are it's a steel bolt with negligible friction and mass. The final scenario for this saw would be cutting a standard 4x4 piece of wood with no kickback. The red dot labeled with '**A**' in the diagram signifies the point of tensile stress throughout the bolt. The tensile stress is focused through the threads as they are the point of highest stress in a bolt. Below the diagram are the calculations for the resulting tensile stress. The highest stress is the tensile stress at the base of the threads with a value of 337 MPa.

Free Body Diagram and Forces:



Max Stresses:

Tensile Stress Factor @ A:



Calculations of Stresses:

Tensile Stress:

$\sigma b = (CP+Fi) / At =$ 337 MPa

$Fi = 0.75AtSp$ = 49 kN, Since it's a nonpermanent connection

$C = kb/ (kb+km)$ = 0.312

P = 390 N from tensile force

To find Fi, we must find At and Sp in tables 8-2 and 8-9, respectively. For proof strength, my bolt had no markings, so I chose the imperial grade bolt with highest strength with no markings due to the assumption that it should be a strong bolt as its job is to hold the blade to the saw. Converted both to metric units to stay consistent.

At =145.81 mm^2
Sp = 65 kpsi = 448 MPa

To find C, we had to find the values kb and km. At was found above. E, A, and B are found in table 8-8 which are the values stated below.

kb =(Ad*At*E) / (Adlt+Atld) = 2917605
km = E·d·A·exp(Bd/l)= 6434532
Ad = 197.93 mm^2

d = 15.875 mm

E = 207 GPa, A = 0.787, B = 0.628

lt = 8.688 mm
ld= 2.25 mm
l = 10.938 mm

Assumed/Chosen Material and Justification:

Factor of Safety Calculations:

I picked cold-drawn medium carbon steel SAE 5 as the material for the bolt, as it has minimal markings, and it is a common cheap bolt to be used in a mechanical setting. Below are also the stress and fatigue factors of safety, and their preceding calculations. I went with Goodman for fatigue factor of safety because it is the most common theory, and it was the one recommended in the textbook to be used.

Below are the two equations used to find static and fatigue factors of safety. For static FOS, all of the values were previously found in calculating the tensile stress, so those values can just be plugged in. For fatigue FOS, I had to find the values of endurance strength and ultimate tensile strength. Using these values, I was able to calculate the remaining σi, σa, and σm.

Static FOS:

$$n_p = \frac{S_p A_t}{CP + F_i}$$

N_p = 1.33

Fatigue FOS:

$$n_f = \frac{S_e(S_{ut} - \sigma_i)}{S_{ut}\sigma_a + S_e(\sigma_m - \sigma_i)}$$

N_f = 0.70

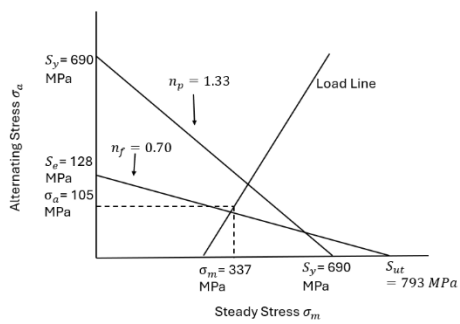
Se = 128 MPa, Sut = 793 MPa
σi=Fi /At
σa = 122 MPa, σm = 337 MPa

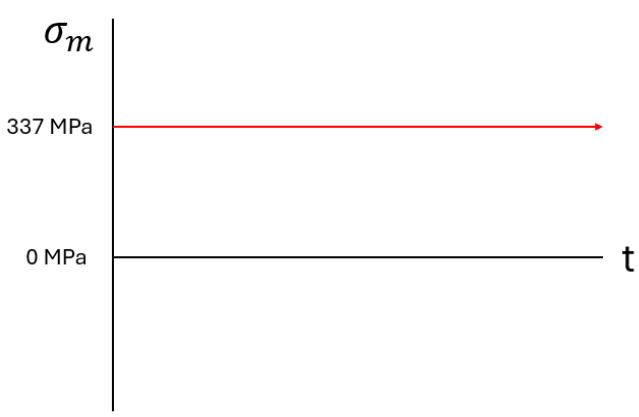
Stress Over-Time and Reliability:
Cycles for fatigue factor of safety until failure:

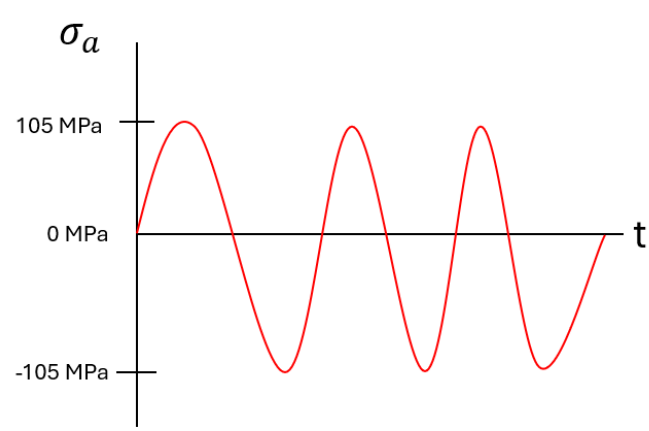
$$N = \left[\frac{\sigma_{ar}}{a}\right]^{\frac{1}{b}} = 227{,}000 \text{ cycles}$$

σar = 182.17 MPa, a = 3377, b = -0.237, f = 0.83

The bolt should be able to go through about 227,000 cycles until fatigue failure occurs. Below are the stress over-time graphs along with the load line graph evaluating the factors of safety.



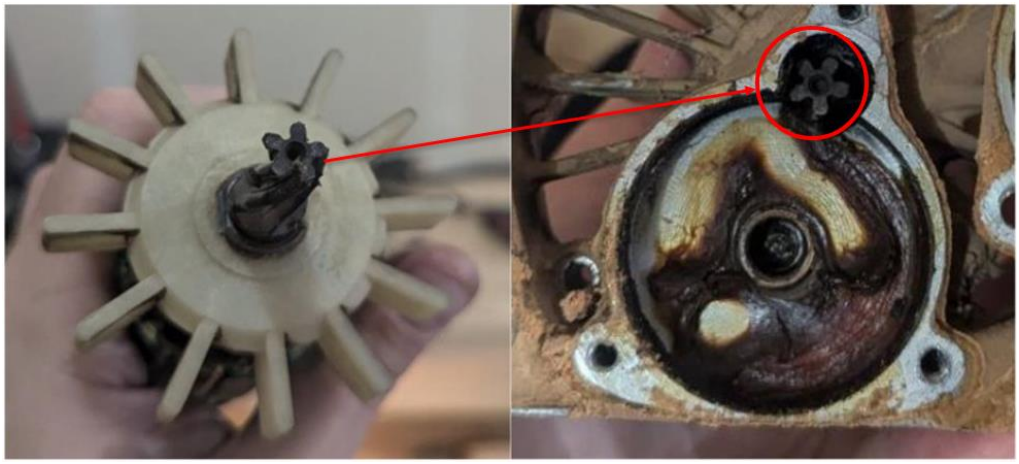




ME383_BoltCalcs

Driving Gear:

The driving gear was chosen as the gear to analyze as this gear would be the simplest gear to obtain torque values for since it is the gear closest to the motor. This component is one of the most critical components in the product as it serves as the primary interfacing component for power transmission—the saw cannot cut without sufficient power. This gear also affects the gear ratio in the transmission which directly affects the speed at which the blade spins—the blade must cut through common materials in a reasonable amount of time.

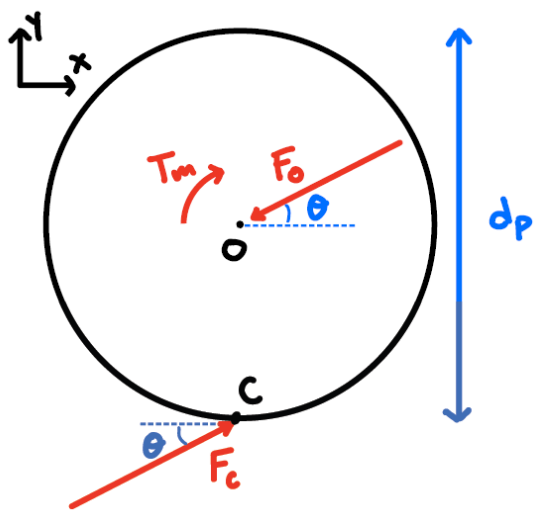


The driving gear outside and inside the housing

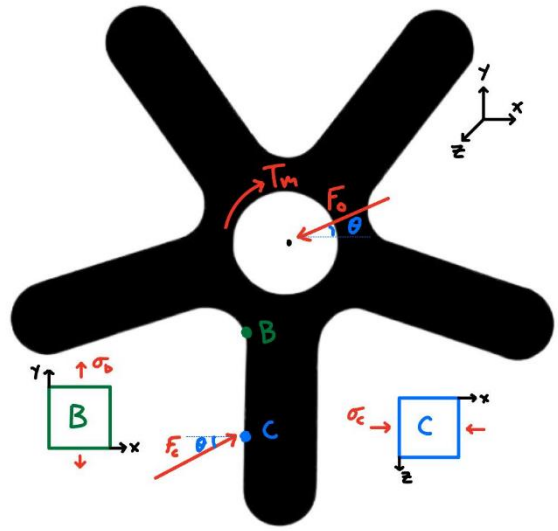
Some key assumptions and simplifications for analysis:

- The gear will be modeled as a spur gear with a profile obtained from the edge face.
- The teeth are perfectly formed, smooth, and rigid and of a commercial quality.
- The gear has a negligible mass.
- Power transmission is lossless between meshed gears and load is uniform.
- Bending moments due to transmission forces are negligible—constant pitch diameter.
- Driving gear and driven gear are made of the same material.
- Gear teeth are uncrowned—therefore linear tooth contact can be assumed.
- Gear rotation is perfectly aligned with the motor shaft which is perfectly balanced.
- Stress is linear with transmitted load.
- Gear will be analyzed for a 10^7 lifespan and for a 99% reliability.
- Operating temperatures will not exceed 250°F.

Stresses and safety factors calculated using ANSI/AGMA Standards 2001-D04 and 2101-D04 (US customary units). Variables, equations, and necessary tables referenced from *Shigley's Mechanical Engineering Design* 10th edition unless otherwise specified. Equations and detailed calculations not included in report in interest of effective results communication. A spreadsheet with these calculations will provide more detail.



Simplified gear free body diagram



FBD of gear overlaid on simplified drawing of gear.

Point B is the location of highest bending stress. This point lies at the base of the teeth, approximately where the tooth meets with the dedendum diameter of the driving gear.

Point C is the location of highest contact stress. This point lies where the two teeth contact each other, approximately located at the pitch diameter of the driving gear for this component.

Free body diagram force analysis results:

Variable	Description	Value	Units
d_p	Pitch diameter	0.35	in
θ	Pressure angle	20	deg
F_o	Force on gear shaft	16.2	lb-f
F_c	Tooth contact force	16.2	lb-f
T_m	Motor torque	13.5	in*lb-f

Stress analysis results:

$$\sigma_b = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_B}{J}$$

$$\sigma_c = C_p \sqrt{W^t K_o K_v K_s \frac{K_m}{d_p F} \frac{C_f}{I}}$$

AGMA Stress Equations (US customary units)

Variable	Description	Value	Units
σ_b	Bending stress	423	psi
σ_c	Pitting resistance (contact stress)	-35,200	psi
σ'	Von Mises Stress	34,700	psi

Calculations and detailed descriptions of variables contained in [linked spreadsheet](#).

Assumed gear material: Class 30 ATSM A48 gray cast iron.

Reasoning: This is a cheap circular saw. The manufacturer would seek to minimize cost through production at scale. A complex part such as this unique gear that is formed onto the motor armature would be unlikely to be machined. As such, a low grade cast iron seems a reasonable material to assume.

Safety factor calculations:

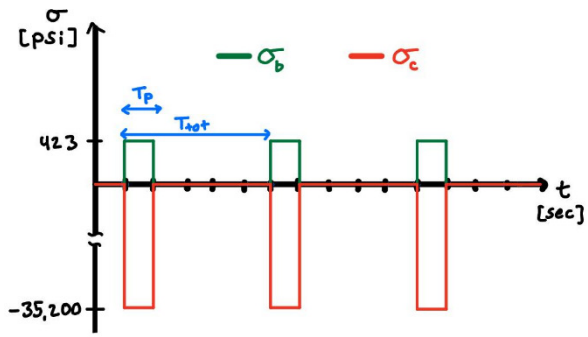
Variable	Description	Value	AGMA Equation (US Customary Units)
S_F	Safety factor against bending fatigue	4	$S_F = \frac{S_t Y_N}{\sigma_b K_T K_R}$
S_H	Safety factor against pitting fatigue	20	$S_H = \left(\frac{S_c Z_N C_H}{\sigma_c K_T K_R}\right)^2$

Calculations and detailed descriptions of variables contained in linked spreadsheet.

Stress over time:

Using rotational dynamics calculations for the pinion. Where $w_{tot} = n_p * \frac{2\pi}{60}$, T_tot = $\frac{2\pi}{w_{tot}}$, and $T_p = \frac{T_{tot}}{N_p}$, where N_p is the number of teeth on the pinion.

Stress vs Time



The stress over time graph is analyzing the stress on a single tooth on the pinion. Note, T_p is the period over which the stress is applied and T_{tot} is the total rotational period of the pinion. These periods would correspond to the cycle times used for fatigue analysis. Since the pinion has 5 teeth, the stress will be assumed to act on a particular tooth the entire time it is in contact with a tooth on the gear. For this analyzed tooth, the cycle time is about 40 ms.

The real world conditions may vary and the rising and falling edges would not have an infinite slope. The shape of this graph is the consequence of the assumptions made in analyzing the gears using AGMA methods. **Key to this analysis is the assumption that the gear will have a 10^7 cycle lifespan**.

Stress over time interpretation:

Variable	Description	Value	Units
σ_{min}	Minimum stress	423	psi
σ_{max}	Maximum stress	-35,200	psi
σ'	Mean stress (Von Mises)	34,700	psi

Note: σ_{max} and σ_{min} are not acting on the same planes, and the stress values are obtained under the assumption that stress is instantaneously toggling from its maximum value to zero, as such, a mean stress value does not give a meaningful value. It is perhaps more appropriate to analyze the von mises stress in this scenario.

Link to spreadsheet containing all relevant calculations for gear analysis.

Spring:

Free Body Diagram and Forces:

Image: Free Body Diagram of Extension Spring

k = constant, pounds of load per inch of deflection	5.821	dx=change in distance for the spring in inches	2
G = modulus of rigidity of spring material, Mpsi	11.0	Fs in pounds (lbs)	11.643
d = wire diameter, inches	0.05		
N = number of active coils, which is the number of coils subjected to flexure	75.0		
D = mean coil diameter, inches = Outer Diameter - Wire Diameter	0.270		

Image: Excel Calculation of Spring Constant (K) and Spring Force

Final Assumptions:

One assumption that I made is that the spring is unpeened; this assumption can be made because the saw and its materials are relatively cheap and inexpensive, so it is most likely that the manufacturer didn't take the extra steps in strengthening the spring. The mass of spring is also negligible. Finaly, it was assumed that spring was not under any pre-existing tension, which aligns with standard scenarios in textbook examples of similar spring mechanisms. This assumption allowed us to focus on spring's response to typical operating conditions without factoring in additional pre-load forces, which would add complexity to the analysis.

Max Stresses:

@ Point A: Maximum tensile stress, due to bending and axial loading

@ Point B: maximum torsional stress

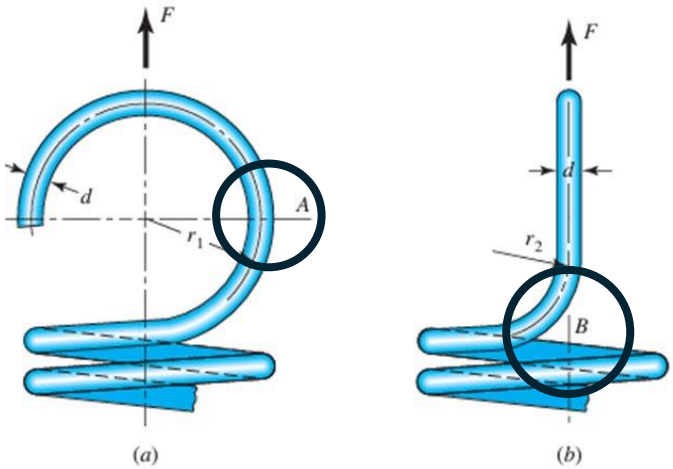


Image: Stress Locations

Calculations of Stresses:

σA	149.2		τB	180.07
(Ka)	1.15		(Kb)	1.125
C1	5.4		C1	7
r1	0.135		r2	0.175

Image: Excel Calculation for Stresses at Points A & B

Assumed/Chosen Material and Justification:

The material is a music wire, 0.80-0.95C, it is the best, toughest, and most widely used of all spring materials for small springs. Music wire has the highest tensile strength and can withstand higher stress under repeated loading than any other spring material. As stated, before this spring is unpeened due to the cheapness of the product.

Stress Over-Time and Reliability:

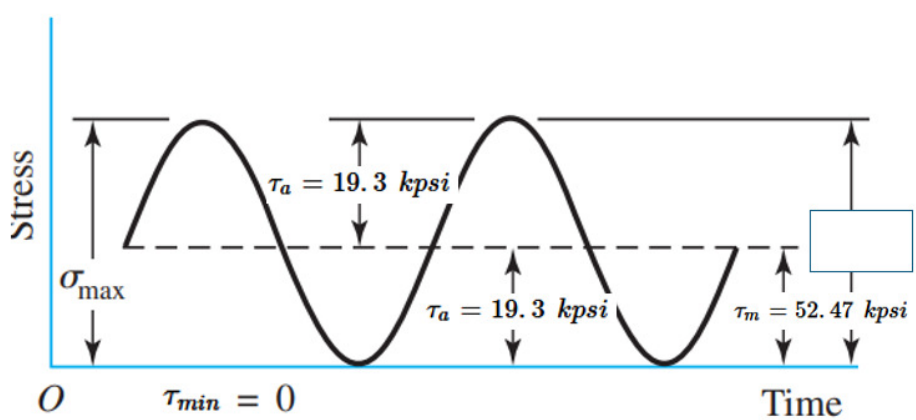


Image: Stress vs. Time graph

Using the adjusted Gerber-Zimmerli equations, the calculated value for the Fatigue Factor of Safety is 1.8 and the Static Factor of Safety is 2.66. The equations are derived from the textbook in chapter 10.

$$n_f = \frac{1}{2} * \left(\frac{S_{sa}}{\tau_m}\right)^2 * \left(\frac{\tau_a}{S_{se}}\right) * \left[-1 + \sqrt{1 + \left(\frac{2 * \tau_m * S_{se}}{S_{su} * \tau_a}\right)^2}\right]$$

Equation: Fatigue Factor of Safety

$$n_s = \frac{S_{sy}}{\tau_s}$$

Equation: Static Factory of Safety

Unpeened:

$$S_{sa} = 35 \text{ kpsi } (241 \text{ MPa}) \qquad S_{sm} = 55 \text{ kpsi } (379 \text{ MPa}) \qquad (10\text{–}28)$$

Peened:

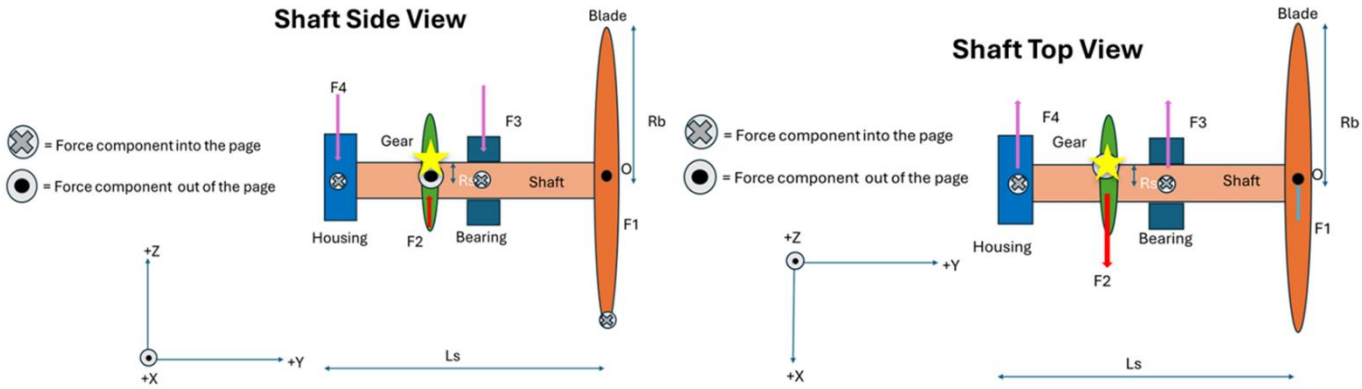
$$S_{sa} = 57.5 \text{ kpsi } (398 \text{ MPa}) \qquad S_{sm} = 77.5 \text{ kpsi } (534 \text{ MPa}) \qquad (10\text{–}29)$$

Photo: Zimmerli Data

Shaft:

ME 383 Spreadsheet Shaft

Free Body Diagram and Forces:



Images: Shaft Side and Top View with Star at Point of Highest Stress

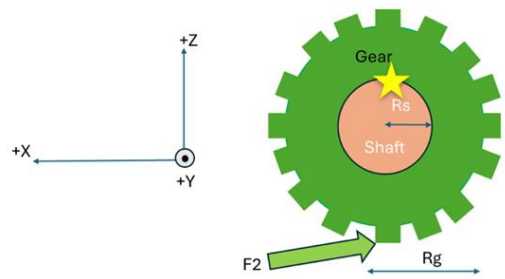


Image: Shaft Cross-Section View With Point of Highest Stress

			Units
Force of Shaft from Blade(x)	F1x	20.000	lbs
Force on Shaft from Blade(z)	F1z	0.000	lbs
Force total on shaft from blade	F_1 = Sqrt(F1x^2 + F1z^2)	20.000	lbs
Force on shaft from Gear (x)	F2x	85.000	lbs
Force on shaft from Gear (z)	F2z	30.000	lbs
Force total on shaft from gear	F_2 = Sqrt(F2x^2 + F2z^2)	90.139	lbs
Force on shaft from Bearing (x)	F3x	23.750	lbs
Force on shaft from Bearing (z)	F3z	22.500	lbs
Force total on shaft from bearing	F_3 = Sqrt(F3x^2 + F3z^2)	32.716	lbs
Force on Shaft from Housing (x)	F4x	41.250	lbs
Force on Shaft from Housing (z)	F4z	7.500	lbs

Image: Calculated Values of all Forces shown in Free Body Diagrams

Final Assumptions:

The final assumptions for our shaft are as follows:

Our shaft has a constant diameter. There is no significant temperature variance anywhere in the shaft. All forces are acting as point forces. Gravitational Forces from the shaft are negligible. The blade force acts parallel to the shaft. The gear acts at a 20-degree angle to the shaft. The final scenario we are analyzing is cross-cutting a 4x4 beam of solid wood.

Max Stresses, Calculated Safety Factors, and Stress Calculations:

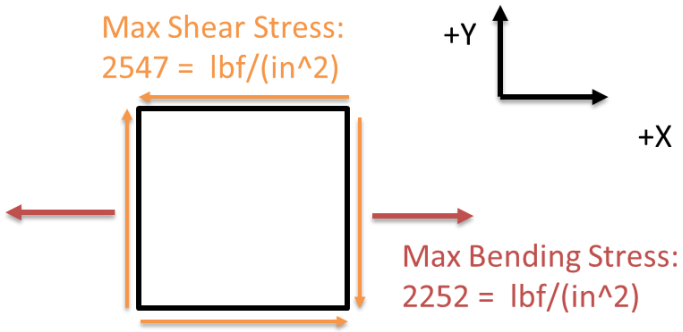


Image: Stress Element Detailing Max Shear and Bending Stress

Yield Factor of Safety:	N_y = S_y / S_Maxp	11.790	Unitless
Fatigue Factor of Safety (Goodman)	N_f = (S_a/S_e + S_m / S_ut)^-1	4.278	Unitless

Image: Calculated Yield and Fatigue Factors of Safety

Value of Q	q_1	0.6	unitless
Value of Qs	qs_1	0.7	unitless
Value of Kt	Kt	1.7	unitless
Value of Kts	Kts	1.4	unitless
Correction Factor for Bending	K_f=1+q_1 * (kt-1)	1.420	
Correction Factor for Torsion	K_fs=1+qs_1 * (kts-1)	1.280	

Image: Calculated Correction Factors from K_t(s) and Q(s)

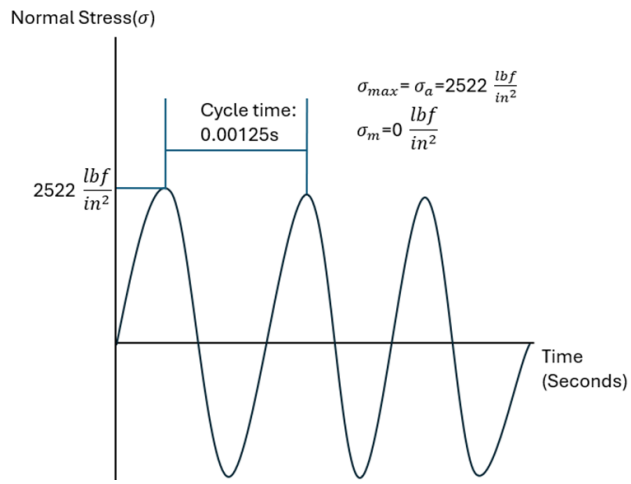
von mises stress:	S_prime =SQRT((K_f)S_maxp)^2+3 * (K_fs	6688.310	lbf / in^2
Alternating Stress:	S_a =SQRT((K_f)S_ab)^2+3 * (K_fs Ts_a)^2	3581.656	lbf / in^2
Mean Stress:	S_m =SQRT((K_f)S_mb)^2+3 * (K_fs Ts_m)	5648.471	lbf / in^2
Max Stress:	S_maxp = S_prime	6688.310	lbf / in^2

Image: Stress Calculation Final Values

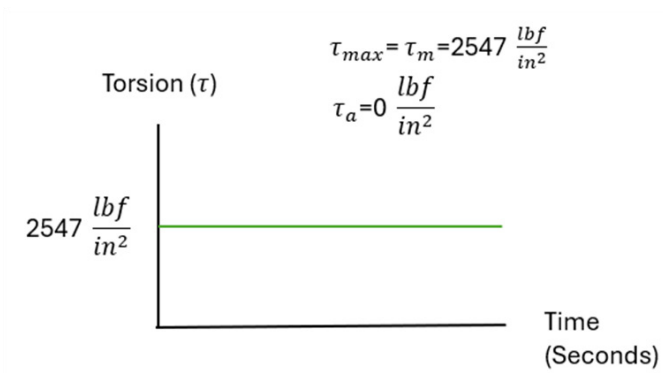
Assumed/Chosen Material and Justification + Failure Theory Analysis:

I decided to use a 1045 cold-drawn carbon steel. The reason I've used this material is because the actual shaft was not given any actual indication of what the material is. I also know that 1045 cold drawn carbon steel is a very cheap material, and it has a high strength and toughness, which makes is a great material for shafts. Additionally, it is by far the most common material used in drive shafts so I figured it would be a great material for this saw. For the fatigue analysis, I decided to use Goodman, since it's standard for shafts and is the failure theory we use in this class on shafts. This works well for shafts because it is a conservative theory, which is useful since if a shaft breaks, the entire saw will completely break and potentially harm the user. For the static analysis, I used Von Mises since it is ideal for Ductile Materials and is pretty much the standard for static analysis on shafts.

Stress Over-Time:



Images: Stress Over Time Graphs For Normal Stress



Images: Stress Over Time Graphs For Shear Stress

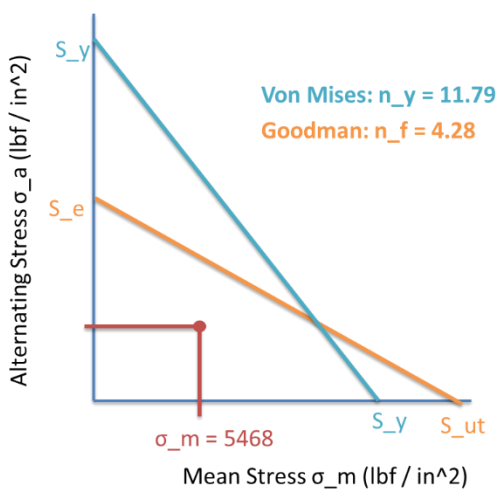


Image: Sm and Sa plot for Yield and Fatigue

Evaluation and Design Recommendations:

Overall, the circular saw components have relatively low factors of safety, in some cases being less than one. This may indicate that the makers of the saw prioritized cost reduction, while ensuring that the saw met reasonable minimums in regards to safety and lifespan. It is clear from even an initial look at the saw that the materials put into the saw were not top quality. One example of potential cost cutting is with the blade retention bolt. The blade retention bolt has a fatigue factor of safety below one, which indicates that it will not have an infinite lifespan. This seems reasonable given the fact that blade changes will occur at such a low frequency that other components would be likely to fail first, with the number of cycles before failure sitting at around 200,000 cycles. Given the low purchase cost of the saw, it seems reasonable to assume that the manufacturer was not prioritizing a long lifespan.

On the other hand, some of the saw components were designed to a higher standard as failure in some components could be disastrous for the customer and the company's reputation. An example, is with the drive shaft. We believe the makers of the saw decided to go with the diameter and material used inside the shaft to ensure a balance between life span, safety, and cost. The drive shaft was made from 1045 cold drawn carbon steel, which is one of the cheapest steels on the market. Next, they decided to use a slightly thicker diameter than needed. This was likely chosen to give some room for error in regards to engineering calculations and manufacturing. It is also quite obvious that the designers were acutely aware the shaft breaking could have dire consequences, with potential for loss of life and getting the company in a lawsuit. Due to this, the engineers must have decided to go with a middle ground of cheap material and thick diameter at the point of max stress, which ensured a low cost while keeping a high factor of safety.

Suggested component revisions and improvements:

Driving Gear:

1. **Material**: Use class 20 ATSM A48 gray cast iron instead of class 30.

   Variables affected:
   - HB (Brinell Hardness) reduced.
   - S_c (AGMA surface endurance strength) reduced.
   - S_t (AGMA bending strength) reduced.

   Reasoning: Easier to work with and ~ 6% cheaper (when compared using this website).
2. **Geometry**: Reduce tooth length by reducing addendum diameter from 0.5 in to 0.4 in.

   Reasoning: 20% reduction in radius results in 36% reduction in volume ($V = \pi r^2 h$) which should correspond to a 36% reduction in material cost.

Gear revision effects

Unit	Class 30, d_A = 0.5 in	Class 20, d_A = 0.4 in
S_F	20	21
S_H	4	2.2
Relative cost	100	60

Calculations done using properties obtained from the textbook input into gear analysis spreadsheet.

Blade Retention Bolt:

1. **Material:** Use SAE Grade 6 or higher bolt instead of SAE Grade 5.

   Reasoning: Higher strength bolts will have higher tensile strength, endurance strength, and proof strength which will increase safety factors.
2. **Geometry**: Decrease the diameter of the bolt from 5/8 in. to ½ in.

   Reasoning: This will decrease the preload force due to a smaller tensile stress area (A_t) which will increase the safety factor.

Bolt revision effects

Unit	5/8 in. Diameter	½ in. Diameter	Adjusted Cycles
n_f	0.70	0.96	827,000

Blade Guard Return Spring:

1. **Shot Peening**: Shot peeing is a cold working process used to impart compressive residual stresses onto the surface of the spring, which results in a modified mechanical property.

   Reasoning: This increases the S_{se} and S_{sm} values ultimately changing the Gerber ordinate intercept for shear (S_{se}). This affects the Fatigue Factor of Safety from 1.8 to 2.9 and the Static Factor of Safety remains the same.

	Unpeened	Peened
n_f	1.8	2.9

2. **Material**: The second change I suggest is using 302 Stainless Steel for the spring instead of using the music wire.

   Reasoning: The 302 stainless steel spring material is approximately 36% cheaper according to one supplier. 302 stainless steel also able to handle a higher load and is rated for more lbs/in. The change in material also changes the Fatigue Factor of Safety changes to 1.48 which is still in the reasonable range.

302 Stainless Steel Extension Spring - P/N 2727

Part Number	2727
Outside Diameter (in.)	0.125
Inside Diameter (in.)	0.097
Overall Length (in.)	0.750
Wire Diameter (in.)	0.014
Rate (lbs./in.)	1.01
Load (lbs.)	0.63

6 Piece Minimum Order Quantity	
1-49 piece price	$2.96
50-99 piece price	$1.74
100-249 piece price	$1.51
250-499 piece price	$1.31
500-999 piece price	$0.90
1000-1999 piece price	$0.49

Image: 302 Stainless Steel Spring Material

Drive Shaft:

1. **Material**: The first change I'm planning to make is to change the material from a 1045 cold drawn carbon steel to a 1018 cold drawn carbon steel.

   Reasoning: I've decided to change to a cold drawn 1018 carbon steel because this only changes the calculated fatigue factor of safety from around 4 to 3.2 which is still plenty to keep the shaft in good shape. Additionally, the 1018 cold drawn steel is about 20 percent cheaper than the 1045 cold drawn carbon steel, which makes this a great decision overall.

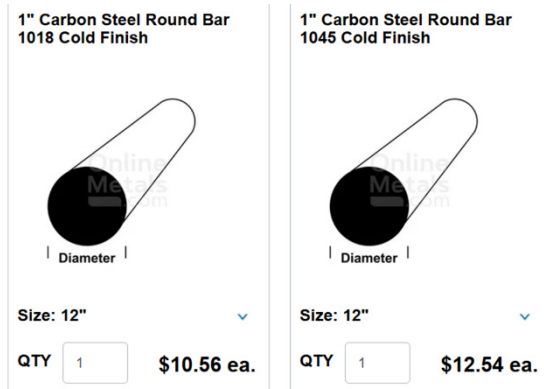


Image: 1018 Cold Drawn Carbon steel vs. 1045 Cold Drawn Carbon Steel price

2. **Geometry**: Reduce the maximum shaft diameter from 0.5 inches to 0.4 inches.

   Reasoning: Using current fatigue factor of safety with the current diameter, we calculate a safety factor of 4. If we reduce the diameter from 0.5 inches to 0.4 inches, we will be able to substantially reduce our cost of making the shaft by using less material, and we will still end up with a fatigue factor of safety of 2.

Fatigue Factor of Safety (Goodman)	N_f = (S_a/S_e + S_m / S_ut)^-1	4.278	Unitless
Fillet Radius	r_1	0.02	in
Smaller Diameter	D_1	0.25	in
Larger Diameter	D_2	0.5	in
Fillet Radius to Diameter Ratio	r/d	0.08	unitless
Larger to Smaller Diameter Ratio	D_2/D_1	2	unitless
Value of Q	q_1	0.6	unitless
Value of Qs	qs_1	0.7	unitless
Value of Kt	Kt	1.8	unitless
Value of Kts	Kts	1.4	unitless

Fatigue Factor of Safety (Goodman)	N_f = (S_a/S_e + S_m / S_ut)^-1	2.088	Unitless
Fillet Radius	r_1	0.02	in
Smaller Diameter	D_1	0.25	in
Larger Diameter	D_2	0.4	in
Fillet Radius to Diameter Ratio	r/d	0.08	unitless
Larger to Smaller Diameter Ratio	D_2/D_1	1.6	unitless
Value of Q	q_1	0.6	unitless
Value of Qs	qs_1	0.7	unitless
Value of Kt	Kt	1.7	unitless
Value of Kts	Kts	1.4	unitless

Images: Calculated Fatigue Factor of Safety Before and After Diameter Change